Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/19/2005
PI Name:	Pometto, Anthony Ph.D.		
Project Title:	NASA Food Technology Commercial Space Center		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	50011-1061	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:	Olson, Dennis		
COI Name (Institution):			
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Task Description:	The mission of the Food Technology Commercial Space Center (FTCSC) is to lead a national effort to develop foods and food-processing technologies that enhance space missions and advance commercial food products through cooperative efforts with NASA scientists and technologists, commercial companies, and academic researchers. In addition to the $2.0 million in cash and in-kind activities contributed by the center's twenty-eight commercial partners, NASA FTCSC research funds are leveraged with affiliate faculty research programs by purchasing equipment that will enable their research program to perform a NASA FTCSC research function. During FY 2003/04, NASA FTCSC-funded equipment and seed grant requests for center affiliate faculty totaled $30,650 and $18,000, respectively. With these equipment purchases, NASA FTCSC gained an additional $3.2 million of non-NASA funds in research efforts with a five-year total of $9.6 million from a total NASA FTCSC investment of $375,000.		

Rationale for HRP Directed Research:

NASA FTCSC affiliate faculty and commercial partners are working to develop foods and food-processing technologies that benefit NASA and the public. The Center's commercial partners are developing products for space and terrestrial applications.
Advanced Analytical Technologies, Inc. (AATI) developed a testing kit that allows the identification of contamination of beef carcasses by E. coli O157:H7 in 18 hours rather than the 24-48 hours required by other testing methods.
Analytik Jena AG conducted five studies of the relationship between food quality and health: 1) antioxidant levels in soybeans and their potential effect on quality, 2) the effect of light on the antioxidation potential of milk products, iron status, and oxidation potential in humans, 3) the effect of extrusion on the antioxidant potential of soy and changes in vitamin E, 4) the effect of phytate and isoflavone in soy on antioxidant content in human plasma, and 5) change in antioxidant potential in athletes under stress.

Ajinomoto Co., Inc. tested many variations of cooking rice, aseptic packaging of cooked rice, storage temperatures, and reheating pre-cooked rice. The tests have found a method that produced rice of good eating quality and long shelf life that requires 90 percent less water than conventional steaming. To prepare the pre-cooked rice for serving, the water must be 185 degrees F, which the ISS cannot supply. More research on final cooking is needed.

Arla Foods amba developed fourteen safe, extended-life products from milk. These include three enriched drinks, two enriched yogurts in powdered form, five spoonable yogurts, and two calcium-enriched protein "bites." Arla has almost finished development of a thermally stabilized yogurt and a dehydrated whole milk.

Biosteam Technologies, Inc. A test of chlorine dioxide conducted by Fresh from Texas, a food handling company, demonstrated that chlorine dioxide has potential as a sanitizer and disinfectant for fresh fruits and vegetables.

Biotecgen s.r.l. is developing DNA arrays with sequences able to discriminate between lactobacillus species. The methodology for PCR amplification and DNA labeling is in place. The DNA array is called the Lacto-chip. Results will be published soon. Biotecgen has developed a hydratable box that contains salad seeds treated for longevity. Antifungine proteins preserve the seeds from molds.

California Dried Plum Board researched the suppression of foodborne pathogens and retarding lipid oxidation in cooked meat products by the addition of dried plums; prebiotic benefits and suppression of less-desirable microbes in the human intestinal flora by dried plum consumption.

East West Medical Research Institute/Mariani Packing Company/Pharmachem Laboratories, Inc. developed a novel 100 percent fruit product that allows enrichment with nutraceuticals. The first product, due for release in Fall 2004, is a children's vitamin. The second, also due for release in Fall 2004, is enriched with calcium citrate to counter osteoporosis.

Heritage Fare, Ltd. Adding a novel fat substitute to irradiated ready-to-eat (RTE) hamburger patties raised their levels of moisture and carbohydrate and lowered their lipid level. Adding plum extract improved the stability of the patties during 30 days of storage and made them juicier. Both treatments had less effect on the quality of cooked patties than processing treatments.

HR Mtn. Sun, Inc. Using the technology developed to make the original pear bar, HR Mtn. Sun, Inc. has developed pear bars that offer enrichment with calcium, pear pieces, flax, or pea protein. Working with the USDA, HR Mtn. Sun is working on versions that include potato or soy. The government of Malaysia is interested in developing a mango bar.

Hy-Vee, Inc. conducted a marketing campaign with print advertisements and in-store signs to promote awareness of Triple "F", Inc.'s soy chili and its availability in Hy-Vee supermarkets. Hv-Vee also completed its process for introducing new products.

Research Impact/Earth Benefits:

Iowa Soybean Promotion Board/Soyfoods Council. Four research teams (Iassonova, Waniska, Wilson, and Myers) have developed soy products. Two processes were developed to make cheese-flavored products from a mix of soy protein and milk fat. Tortillas with five percent soy flour were developed. Snack chips with high nutritional value were produced from a mixture of wheat and okara. A soy-based adhesive is in technology transfer.

Kemin Americas, Inc. developed four formulations based on rosemary extract that extend the shelf life and stability of many products including processed meats, sauces, baking mixes, tortillas, fats, oils, nuts, salsas, beverages, and dry ingredients. Kraft Foods, Inc. supported research on improved packaging materials and bone health. Packaging of lighter weight and greater strength has been achieved with the use of nano-filler reinforcement. A study of nutrients with potential benefits for bone health is underway.

Maytag Corporation. Continued research on a refrigerator designed to extend the shelf life of fresh fruits, vegetables, and meats; and an oven designed for energy efficient heating of foods.

MCC, s.a. (Meiners Commodity Consultants) researched microencapsulation and controlled release of vitamins and minerals.

Microbial-Vac Systems, Inc. Microbial-Vac Systems has developed a small, low-power wet-vac that samples surface contamination with high efficiency and provides an integral container for safe transport and centrifugation that saves time and expense.

Microcide®, Inc. Three antimicrobial preparations were tested on tomatoes inoculated with nalidixic-acid-resistant Escherichia coli 0157:H7 or Salmonella spp. Washing with water, a solution of sodium hypochlorite, or hydrogen peroxide produced reductions in pathogenic bacteria and increases in shelf life. Washing with PRO-SAN (Microcide's proprietary cleaner) produced significantly greater reductions in pathogenic bacteria and spoilage of tomatoes in storage.

Monsanto. Improving food production and composition through biotechnology and genetics on Earth and on astronauts' diets in space. Explore future crop production on planetary outposts.

MTI BioTech, Inc. tested the potential of two products to improve healing and maintain and restore muscle. One product, Juven®, is commercially available. It combines amino acids and HMB (ß-hydroxy-ß-methylbutyrate, a metabolite of leucine). The other product is HMB®. It is commercially available. Both helped maintain or increase muscle in tests with cancer patients and elderly women.

	ORBITEC has prototyped a container that stows flat and stacks, expands as needed, and carries a high- and medium-intensity lamp for growing plants. Pactech tested the strength and permeability of flexible film laminates and the strength and tightness of fitments that seal bags and pouches made from the laminates. Pactech has produced bags that are highly impermeable and tightly sealed and that have the potential to extend the shelf life of stored foods to five years and beyond. Pactech also tested thermal pouches that have potential to carry food to Cape Canaveral without refrigeration. Pioneer Hi-Bred International, a DuPont Company has grown suitable experimental soybeans and produced seeds for breeding nurseries and field experiments, from which the first harvest will be made in Fall 2004. Ross Products Division, Abbott Laboratories Inc. Development of a nutritional beverage designed to maximize maintenance of muscle mass and repair of tissue. Triple “F”, Inc. produced two forms of soy-based food that NASA has accepted and is developing seven more products. Prototypes of soy-based “hamburger” and “meat balls” have been sent to NASA.
Task Progress:	The NASA Food Technology Commercial Space Center’s fifth year of operation brought tremendous strides in its mission of developing foods and food-processing technologies that enhance space missions and advance commercial food products. The Center’s commercial partner program increased to twenty-eight partners. Additions to the program include: Corporate Founding Partner Pharmachem Laboratories, Inc.; and Affiliate Partners Biosteam Technologies, Inc., Monsanto, ORBITEC, and Ross Products Division, Abbott Laboratories Inc.. The Center added ten new affiliate faculty members bringing the current total to forty-nine. These faculty members, located at research facilities throughout the world, are addressing almost every critical area of space food technology research. NASA FTCSC hosted its first research symposium, "Developing the Next Generation of Space Food Technologies: Engaging Industry and Academia" on April 21, 2004. More than 100 people attended the event that showcased the past five years of research accomplishments. Attendees included 17 commercial partners, 64 affiliates, seven universities, one collaborating center, one USDA federal laboratory, and three NASA experts. Three NASA FTCSC commercial partners engaged affiliate faculty in NASA-related research projects. Drs. Aubrey Mendonca and Cheryll Reitmeier illustrated the benefits of Affiliate Partner Microcide® Inc.’s PRO-SAN™ over hypochlorite and peroxide surface sanitizers for ISS. Dr. Dong Ahn demonstrated the benefits of the addition of a fat substitute and dried plum extract from Affiliate Partners Heritage Fare, Ltd. and the California Dried Plum Board, respectively, to cooked hamburgers that were freeze-dried or irradiated for ISS. NASA FTCSC commercial partners and affiliate faculty gave seven presentations at the International Conference on Environmental Systems (ICES) in Colorado Springs, Colorado, July 19-22, 2004. NASA FTCSC Director Anthony Pometto and Consultant Charles Bourland gave an update on NASA FTCSC research. Center affiliate faculty Drs. Ahn, Hammond, Irudayaraj, Mendonca, Seetharaman, and Wilson presented their research findings that included overviews of FTCSC-related research by Commercial Partners Analytik Jena AG; Bready International, AB; Heritage Fare, Ltd.; Iowa Soybean Promotion Board/Soyfoods Council; and Microcide, Inc. NASA FTCSC education and outreach activities have again been very successful thanks to the hard work and dedication of Dr. Cheryll Reitmeier, education mission specialist, and Dr. Aubrey Mendonca, outreach mission specialist. Based on the success of Reitmeier’s USDA Higher Education Grant, the Center submitted its first NSF Center for Learning and Teaching (NSF CLT) proposal which proposed to use space food technology as a vehicle to teach science, technology, engineering, and mathematics (STEM) principles to high school and college students. In the past year, Mendonca visited Florida A&M University, University of Florida, and Alabama A&M University. The NASA FTCSC continues to engage food scientists from historically black 1890 Land Grant Universities, with the University of Arkansas at Pine Bluff participating in the NSF CLT proposal. New NASA FTCSC Affiliate Partner ORBITEC collaborated with the Center on a proposal to design a HACCP plan for salad crop production on the International Space Station. ORBITEC secured NASA SBIR funding for this project. In FY 2003/04 there was one commercial success to report from Commercial Founding Partner, Hy-Vee, Inc. Their deliverable is the marketing of an approved NASA FTCSC product as food for astronauts. In December of 2003, Hy-Vee completed a marketing campaign for the astronaut-menu approved soy chili developed by Corporate Partner Triple “F”, Inc. A commercial success, as defined by the NASA FTCSC Advisory Council, is a commercial partner product that is both approved by NASA JSC and commercially available. NASA JSC approved Hy-Vee’s advertisement and Hy-Vee's stores in the Midwest marketed the soy chili.
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